

Backstop Study 2.0

Benchmarking Agentic Cost-Escalation Guardrails

A Redlynr Validation Study · Executive Summary
HDGForge Labs, an HD Gregory LLC company · 2026

What Was Tested

Redlynr is a per-request guardrail API for LLM agent chains. It sits alongside an agent’s tool-calling loop and monitors per-step behavior — cost accumulation, loop patterns, tool fixation, retry storms, and delegation depth — returning a structured pass/slow_down/stop decision with supporting signal evidence on every call. Backstop answers a single question: does Redlynr actually stop the runaway patterns it claims to stop, without intervening on legitimate traffic?

The Verdict

Redlynr’s cost-escalation guardrail was validated across all three tested models and both providers. Across 1,568 trials spanning 28 scenarios, 4 conditions, and 10 domains, C1 and C2 achieved a 100% true positive rate on cost-escalation incidents (Wilson 95% lower bound: 96.3%) and a 0% false positive rate on legitimate workloads (Wilson 95% upper bound: 4.4%). Median guardrail decision latency was 6.4 ms with no statistically significant variation across models. The study also surfaced four genuine findings — all identified and corrected before Phase 2 data collection began.

Results at a Glance

Each condition was evaluated across incident trials (measuring true positive rate) and control trials (measuring false positive rate). All figures reported with 95% Wilson score confidence intervals in the full report.

Condition	Cost-Esc. TPR	FPR (Control)	p50 Latency
C1 — Factory defaults	100% (98/98)	0% (0/84)	6.36 ms
C2 — Tuned policy	100% (98/98)	0% (0/84)	6.43 ms
B — Naive rate limiter	66.3% (65/98)	48.8% (41/84)	N/A
A — No guardrail	0% (0/98)	N/A	N/A

Wilson 95% lower bound on C1/C2 cost-escalation TPR: 96.3%. Wilson 95% upper bound on C1/C2 FPR: 4.4% (observed rate 0%; n=84 control trials). Loop detection validated on 14 genuinely stuck incidents; 100% catch rate at both C1 and C2. Retry-storm scenarios all self-resolved in condition A; retry-storm is not a validated incident category in this study.

Model	Provider	Trials	Repeats	C1/C2 Cost-Esc. TPR	C1/C2 FPR
GPT-5.6 Luna	OpenAI	560	5	100% (35/35)	0% (0/15)
GPT-5.6 Terra	OpenAI	448	3	100% (21/21)	0% (0/9)
Claude Haiku 4.5	Anthropic	560	5	100% (35/35)	0% (0/18)

Terra’s Wilson intervals are wider due to 3 repeats versus 5 for Luna and Haiku. Wilson lower bound (cost-esc. TPR): Luna 90.0%, Terra 84.5%, Haiku 91.5%. Wilson upper bound (FPR): Luna 11.4%, Terra 13.8%, Haiku 11.4%.

How the Study Was Conducted

Two phases, four conditions

Phase 1 — Pre-spend validation gate: four test suites (80 mock trials, 48 behavioral trials, 54 adversarial trials, 9 batch-cost-fix validation trials) required to pass at full count before any inference budget was committed. This gate caught four findings in Redlynr’s implementation before any live model inference ran.

Phase 2 — Full pilot runs: 28 scenarios across 10 domains, evaluated under all four conditions against three models and two providers. 1,568 trials total.

Key methodological decisions

- Mechanism-first: each guardrail check’s actual trigger condition was confirmed via direct test calls before any inference budget was committed. This prevented two significant harness misdesigns during Phase 1.
- Pre-spend gate: all four test suites required to pass at required counts before Phase 2 began. No exceptions.
- Per-tenant isolation: history-dependent checks use Redis state keyed on tenant_id; each trial was assigned a distinct simulated tenant to prevent cross-trial state contamination.
- No-overwrite audit trail: every run is retained in the repository under a numbered directory, including amended runs, so the full history is permanently inspectable.
- Amendments disclosed: four deviations from the original design (model pivot, scenario expansion, batch-loop break fix, repeat asymmetry) are documented in full in the methodology section of the technical report.

Study Findings — Four Corrected Before Phase 2

A secondary goal of this study was to find real defects in Redlynr through mechanism-first, adversarial testing. Four findings were identified and corrected before Phase 2 data collection began.

Finding 1: Cost accumulator not committed to Redis on slow_down decisions (corrected)

When Redlynr issued a slow_down decision, the cost accumulator was not written to Redis — only stop decisions committed it. In chains where slow_down preceded stop, subsequent steps evaluated cost against a stale baseline. Fix: slow_down now commits the accumulator unconditionally. Verified in the batch-cost-fix validation suite (9/9) before Phase 2.

Finding 2: Subagent guardrail stops not propagating to parent loop (corrected)

A stop issued inside a subagent chain was local to that subagent. The parent orchestrator loop continued as though the subagent had completed normally, allowing total cost across parent and subagents to exceed the intended ceiling. Fix: subagent stops now propagate to the parent loop as a structured terminal event.

Finding 3: Batch-loop harness continued processing after stop (corrected)

This is a harness finding, not a product defect. The orchestrator’s batch loop used continue rather than break when a stop was detected on the first item in a parallel tool-call batch. For models that issue multiple tool calls in a single response (GPT-5.6 Luna did this in 38% of incident trials), remaining calls were processed after a stop was already returned. Fix: continue replaced with break. All 1,568 trial records in the final dataset were collected with the corrected harness.

Finding 4: Cost hard stop evaluated after step_cap_warning in check order (corrected)

A chain that had crossed the cost ceiling but not yet the step cap would receive a step_cap_warning rather than a cost hard stop — a lower-severity signal than warranted. Fix: cost hard stop now evaluates first in check order.

What is validated: Redlynr's cost-escalation guardrail stops cost-escalation incidents and stays quiet on legitimate workloads, across GPT-5.6 Luna, GPT-5.6 Terra, and Claude Haiku 4.5. At N=98 cost-escalation incident trials and N=84 control trials with 95% Wilson intervals, this is a statistically powered result on cost escalation specifically. Loop detection is validated on 14 genuinely stuck incidents (crm_loop_01, 100% catch rate). Retry-storm and other incident types where models self-resolved at 100% are not validated categories in this study.

Audit trail: all 1,568 trial-level records, harness code, and methodology documentation are publicly available at
github.com/HDGForge-Labs/backstop-study-2

Reports: github.com/HDGForge-Labs/backstop-study-2/tree/main/reports
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